

Stage 2 Lab: AI Demonstration Copy (Parts A to F)

Part A: Client brief (same at every level)

SmartCare uses spreadsheets and paper records. Staff report duplicate bookings, difficulty finding patient information, inconsistent appointment status and limited appointment history. Management wants a small, maintainable patient, practitioner and appointment system.

Part B: Stakeholders and scope (same at every level)

Stakeholder	Need	Evidence
Reception/admin staff	Book and cancel reliably; find patient information quickly	Duplicate bookings; difficulty finding patient information
Practitioners	See their appointments and availability	Limited visibility of availability
Patients	Correct bookings and easy cancelling	Duplicate bookings; inconsistent status
Clinic management	A small, maintainable system with useful history	Client brief

In scope: patient records and search; practitioner records; creating and cancelling appointments; blocking double-bookings; appointment status; appointment history; practitioner schedule view.

Out of scope: online payment; SMS reminders; facial recognition login; AI treatment recommendations; insurance.

Provisional: exact appointment statuses; history retention period; whether patients book for themselves.

Part C: Functional requirements

ID	Simpler	More advanced
FR-01	Staff can add a new patient with a unique ID.	The system shall create a patient record containing a system-generated unique identifier and mandatory identifying attributes (given name, family name, date of birth).
FR-02	Staff can look up a patient by ID or name.	The system shall retrieve patient records by exact identifier or case-insensitive name match and return all matching records.
FR-03	The system keeps a list of practitioners.	The system shall maintain a practitioner registry with unique identifiers.
FR-04	Staff can book an appointment with a patient, practitioner, date and time.	The system shall create an appointment linking a valid patient identifier and practitioner identifier to a specified date and start time.
FR-05	The system stops a double booking for the same practitioner and time.	The system shall reject an appointment if the practitioner already has a non-cancelled appointment at that date and start time.
FR-06	The system won't book for a patient or practitioner that isn't in the system.	The system shall validate the patient and practitioner identifiers at appointment creation and reject the request with a descriptive error when either is not found.

ID	Simpler	More advanced
FR-07	Every appointment has a status (Booked, Cancelled or Completed).	The system shall assign every appointment exactly one status from a controlled set (Booked, Cancelled, Completed) and reject any other value.
FR-08	Staff can cancel an appointment.	The system shall allow authorised staff to cancel a booked appointment, updating its status to Cancelled.
FR-09	Cancelled appointments stay in the history.	The system shall retain cancelled appointments as records with Cancelled status so they remain visible in the appointment history.
FR-10	Staff can see a practitioner's appointments for a day.	The system shall list all non-cancelled appointments for a specified practitioner and date, ordered by start time.

Part D: Non-functional requirements

ID	Simpler	More advanced
NFR-01	Keep business logic separate from the screens so it is easier to change.	The architecture shall decouple domain logic from presentation so either can be modified without affecting the other.
NFR-02	The booking and cancel rules can be tested without the interface.	Conflict-detection and cancellation rules shall be verifiable by automated unit tests run independently of the user interface.
NFR-03	Every appointment must have a real patient and practitioner.	The system shall enforce referential integrity so that no appointment can be stored without a valid patient and practitioner.
NFR-04	A new receptionist can book an appointment in about 5 steps.	In usability testing, at least 90% of first-time receptionists shall complete a standard booking in 5 or fewer interactions without assistance.
NFR-05	Search and booking are quick for around 1,000 patients.	Search and booking operations shall complete within 2 seconds at the 95th percentile for up to 1,000 patient records.
NFR-06	Only authorised staff can see patient data.	The system shall restrict read and write access to patient data to authenticated, authorised staff through role-based authorisation.

Part E: User stories and acceptance criteria

ID	Simpler	More advanced
US-01	As a receptionist, I want to book an appointment, so patients have a time.	As a receptionist, I want to schedule an appointment by selecting a patient, practitioner and time slot, so that the patient receives a confirmed booking.
US-02	As a receptionist, I want a warning if a practitioner is already booked, so I don't double-book.	As a receptionist, I want immediate conflict feedback when I select an occupied practitioner slot, so that scheduling errors are prevented before they reach the record.
US-03	As a receptionist, I want to search for a patient, so I can find their record fast.	As a receptionist, I want to retrieve a patient record by identifier or name, so that I can access accurate information without delay.

ID	Simpler	More advanced
US-04	As a receptionist, I want to cancel an appointment but keep the record, so the history is right.	As a receptionist, I want to cancel an appointment without deleting it, so that the appointment history remains complete and auditable.
US-05	As a practitioner, I want to see my schedule for a day, so I know who I'm seeing.	As a practitioner, I want to view my daily schedule, so that I can prepare for the patients I will see.

Criteria	Simpler	More advanced
US-02 (failure)	GIVEN Dr Doe is booked at 10:00 on 20 July, WHEN I try to book Dr Doe at 10:00 that day, THEN the system says no and saves nothing.	GIVEN a non-cancelled appointment exists for Dr Doe at 10:00 on 20 July, WHEN a user submits a new appointment for Dr Doe at that time, THEN the system rejects the request, displays a conflict message and persists no new record.
US-03 (failure)	GIVEN there is no patient with ID 999, WHEN I search for 999, THEN the system says the patient was not found.	GIVEN no patient record exists with identifier 999, WHEN a user searches for identifier 999, THEN the system returns an empty result with a "patient not found" message and creates no record.
US-04	GIVEN a booked appointment, WHEN I cancel it, THEN it shows Cancelled and stays in the history.	GIVEN an appointment with status Booked, WHEN authorised staff cancel it, THEN its status becomes Cancelled, the record is retained and it appears in the appointment history.

Part F: AI requirements review - ChatGPT

Prompt used: Act as a software requirements reviewer. Review the SmartCare requirements for ambiguity, inconsistency, missing clarification questions and testability. Do NOT invent new client requirements. For every suggestion, state whether it is based on evidence or is only a question/assumption requiring validation.

Below is a requirements review based **only on the SmartCare case study provided**. I have not added assumed client requirements; where clarification would be needed, I identify it explicitly as a question rather than treating it as a requirement.

The case study states that SmartCare wants a simple application initially supporting **patient, practitioner and appointment management**, intended to replace fragmented manual and spreadsheet processes.

SmartCare requirements review

Area	Issue identified	Evidence / question or assumption	Testability
System scope	"Patient, practitioner and appointment management" is too broad to determine what the system must actually do.	Evidence: The case study explicitly identifies these three areas but does not define their functions. Clarification question: What operations must be supported for each area (e.g., create, view, update, delete)?	Not sufficiently testable until specific operations and expected behaviour are defined.
Patient records	The requirement does not specify what	Evidence: Difficulty locating patient records is identified as an existing problem. Clarification question:	Cannot create complete acceptance tests

Area	Issue identified	Evidence / question or assumption	Testability
	patient information must be stored.	What patient fields are required, and how should patients be searched or identified?	without defined data and search behaviour.
Practitioner management	Practitioner availability is mentioned as a problem, but the required availability functionality is unspecified.	Evidence: The clinic has “limited visibility of practitioner availability.” Clarification question: What information defines availability, and how should it be displayed or maintained?	Partially testable only at the problem level; actual system behaviour is undefined.
Appointment booking	The case identifies duplicate bookings as a problem but does not define when two bookings constitute a duplicate.	Evidence: Duplicate appointment bookings are explicitly identified. Clarification question: What combination of practitioner, date/time, patient or other information should the system use to detect a duplicate?	Not objectively testable until the duplicate-booking rule is defined.
Appointment status	“Inconsistent appointment status information” identifies a problem but does not specify valid statuses or status transitions.	Evidence: Inconsistent appointment status is an existing operational problem. Clarification question: What appointment statuses are required, and under what circumstances can each status change?	Not testable without defined statuses and transition rules.
Cancellation	Manual cancellation is identified as a problem, but the required cancellation process is not described.	Evidence: The current process includes manual cancellation. Clarification question: What should happen when an appointment is cancelled, and who is permitted to cancel it?	Cannot reliably test cancellation behaviour without defined expected outcomes.
Appointment history	“Reliable appointment history” is not defined.	Evidence: Lack of reliable appointment history is an identified problem. Clarification question: What appointment information must be retained as history, and how should users access it?	Not sufficiently testable because “reliable” has no measurable acceptance condition.
Operational reports	“Basic operational reports” is ambiguous.	Evidence: Difficulty producing basic operational reports is identified. Clarification question: Which reports are required and what information must each contain?	Not testable until report types, contents and expected results are specified.
Simplicity	“Simple”, “manageable” and “not a complex hospital information system” are subjective.	Evidence: These descriptions are explicitly used in the case study. Clarification question: What measurable scope or usability	Not directly testable because no measurable criterion is provided.

Area	Issue identified	Evidence / question or assumption	Testability
System users/roles	The case identifies GPs/practitioners and patients, but does not clearly specify who will operate the software.	constraints does management associate with “simple”?	Access-related tests cannot be defined without user roles and permissions.
		Evidence: The organisation provides consultations through healthcare practitioners, while the system is intended to manage patient, practitioner and appointment information. Clarification question: Which categories of staff/users need access to the application?	
Data migration	The system will replace spreadsheet, paper and manual processes, but migration of existing information is not addressed.	Evidence: The current process uses spreadsheets and paper records, and the system is intended to replace those fragmented processes.	This cannot be tested until migration expectations are explicitly established.
		Clarification question: Is existing patient and appointment data expected to be transferred into the new system?	
Appointment history vs cancellation	There may be an unresolved relationship between appointment cancellation and appointment history.	Question/assumption requiring validation: The case says both cancellation and appointment history are concerns, but does not state whether cancelled appointments must remain in history.	Behaviour cannot be tested without clarification.
Reporting scope	There is no stated frequency, format or access method for reports.	Question/assumption requiring validation: The case only establishes a need for “basic operational reports”; it does not establish these additional details.	Insufficient information for objective report acceptance tests.
Iterative development	The case says the prototype will be enhanced in stages, but does not define which requirements belong to which stage.	Evidence: Development is described as iterative, with enhancements in each stage. Clarification question: Which functionality is required at each development stage?	Stage-specific completion cannot be tested without defined scope per stage.

Key inconsistencies

There are **few direct contradictions** in the supplied case study. The larger issue is **underspecification**.

One important tension requiring clarification is:

- The system is described as a “**simple**” and “**manageable**” application, while the stated problems span patient records, practitioner availability, appointments, cancellations, history and reporting.
- **This is not necessarily an inconsistency.** It is a potential scope ambiguity. The client needs to define what “simple” means rather than assuming that particular features should be excluded.

Another potential ambiguity is the relationship between the broad initial scope—patient, practitioner and appointment management—and the operational problems listed earlier. The

case does not explicitly state whether **every listed problem must be solved in the first version**.

Clarification question: Are all seven listed operational problems intended to be addressed by the first version, or are some intended for later iterations?

Testability assessment

The strongest issue is that most requirements are currently expressed as **problems or goals rather than testable system behaviours**.

For example:

“duplicate appointment bookings” — identifies a problem, but not the rule the software must enforce.

“reliable appointment history” — identifies a desired outcome, but does not define what information must be retained or what makes the history reliable.

“basic operational reports” — identifies a desired capability, but does not identify the reports or their required contents.

These statements provide useful **requirements evidence**, but they are not sufficiently precise acceptance criteria on their own.

Highest-priority clarification questions

1. **What specific functions are required for patient, practitioner and appointment management?**
2. **What constitutes a duplicate appointment, and what should the system do when one is attempted?**
3. **What appointment statuses are required and what status changes are permitted?**
4. **What is the required cancellation workflow?**
5. **What information must be retained and displayed as appointment history?**
6. **What are the “basic operational reports” that the client expects?**
7. **What practitioner availability information must be maintained and displayed?**
8. **Who will use the system and what access/permissions does each user type require?**
9. **Does the first version need to address all of the listed operational problems?**
10. **Does replacing the existing paper/spreadsheet process require existing data to be migrated?**
11. **What functionality is expected in each development stage?**
12. **What measurable criteria define the requirement that the application be “simple” or “manageable”?**

These are **clarification questions, not proposed requirements**. The case study does not provide enough evidence to answer them without making assumptions.

Part G - VERIFY the AI Review

- **Accepted (8):** the scope, patient fields, availability, duplicate-booking rule, statuses, cancellation, "reliable" history and reports, and the cancelled-vs-history question. Each is a real gap in the brief.
- **Modified (2):** "simple" needs measurable criteria (only loosely, since the brief is small) and user roles (the brief only says "staff").
- **Accepted as a question only (1):** data migration. Ask about it, but don't make it a requirement.
- **Unverified (1):** which requirements belong to which stage. I can't check that without the full case study.

Good: it made no requirements up.

Weak: it reviewed the case study, not FRs, NFRs or user stories.

Reflection

The AI noticed things I missed. I never asked if the old spreadsheet and paper records needed to be moved into the new system, who would use it, or what counts as a double booking. It didn't make up any requirements. It only asked questions. But it went a bit too far by asking for exact meanings of "simple". It also checked the case study, not my own requirements.

After the review, I would add a question about moving old records. I would also mark two things as "not confirmed yet": the double booking rule, and keeping cancelled appointments in the history. I would say that the numbers in my usability and speed requirements are guesses for now.

Requirements need evidence so only build what the client really asked for. Features nobody asked for waste time and can cause problems, like privacy risks.